Learning Control

Part I: Frequency-Domain ILC Design

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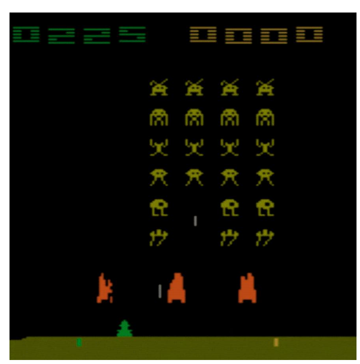
Learning from data

Huge progress in learning

- (video) games: backgammon, chess, go, Atari, starcraft, . . .
- advertising
- generative AI



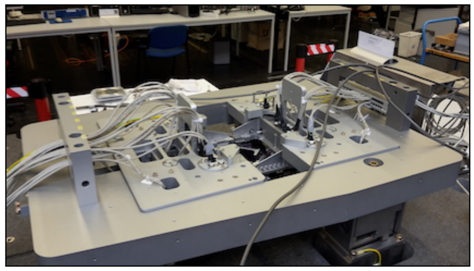
Enabled by

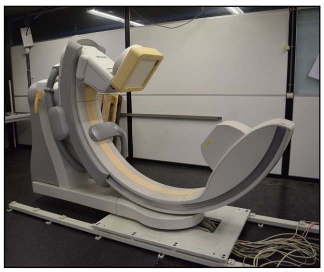
- big data
- ubiquitous computing (e.g., cloud services)

What does learning have to offer for real-world systems?

Learning in Machines

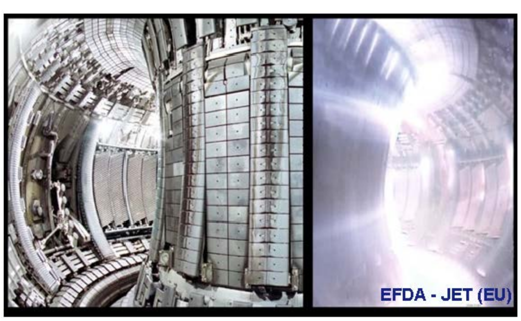
- repeating tasks
- experiments inexpensive
- but still(Oomen 2018*b*)
  - real-time
  - no parallelization
  - time variation
    wear, aging, ...

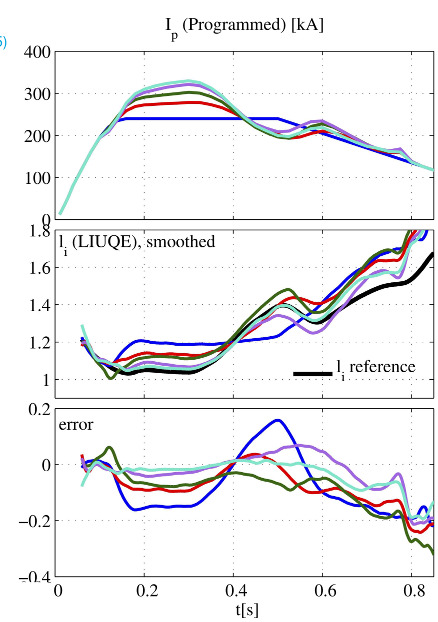




Nuclear fusion: TCV Tokamak, Lausanne(Felici & Oomen 2015)

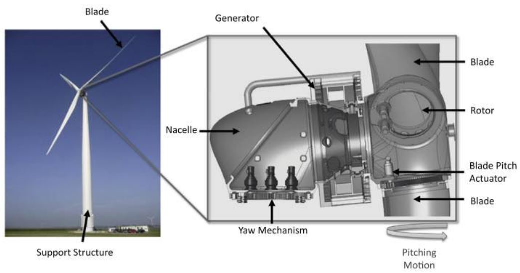
- again repeating tasks
- for successful operation internal inductance l_i must follow a reference
- by adjusting total plasma current I_p
- experiments are rather expensive



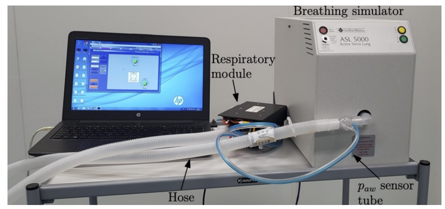


Learning from data

Also energy systems (Navalkar et al. 2014)



and mechanical ventilation (Reinders et al. 2020)

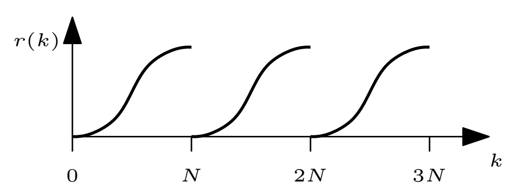


Very similar: repeating tasks in continuous operation (no reset)

Learning from data

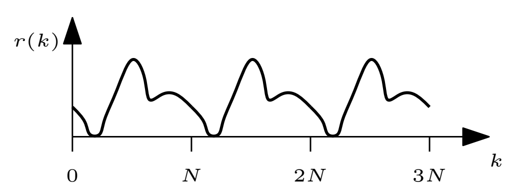
Key properties

- fast: merge model-based approaches and data-driven learning
- safe: provide robustness for safe operation



Main techniques

- Iterative Learning Control (ILC): batch operation
- Repetitive Control (RC): continuous operation



Successfully applied in all previous slides

Learning from data

Historical perspective - Iterative Learning Control

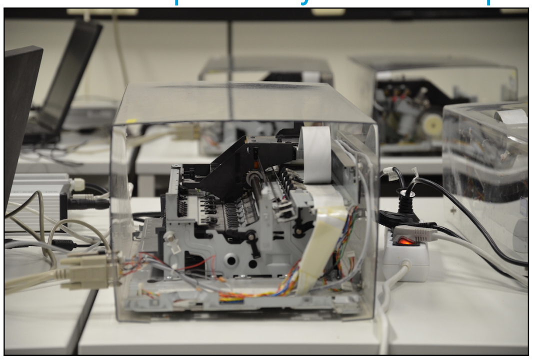
- initial ideas appear in
  - 1967 US patent 3555252: "Learning control of actuators in control systems"
  - 1984 paper (Arimoto et al. 1984): "Bettering operation of robots by learning"
- 1990s-present:
  - well-established theory and design (Bristow et al. 2006) (Bien & Xu 1998) (Gorinevsky 2002) (Moore 1999) (Moore 1993) (Oomen 2018*b*) (Oomen 2020*b*)
  - and many successful applications

Historical perspective - Repetitive Control

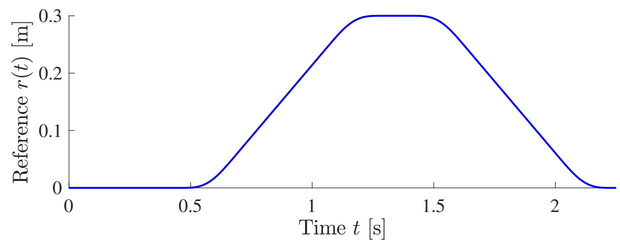
- initial ideas appear in
  - 1976 paper (Francis & Wonham 1976): "The internal model principle of control theory"
  - 1980 paper (Inoue et al. 1981)
- 1990s-present:
  - well-established theory and design
  - and many successful applications

Iterative Learning Control - What Can Be Achieved?

Consumer printer system example

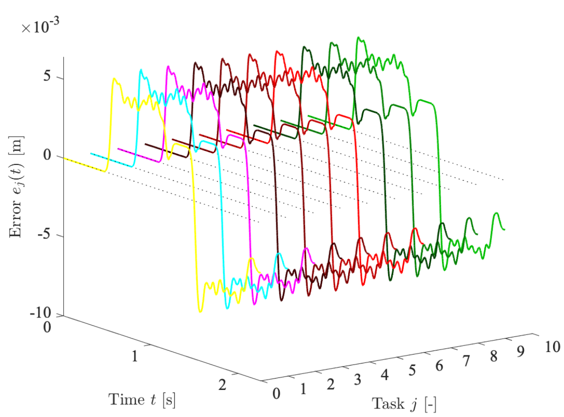


Reference



Iterative Learning Control - What Can Be Achieved?

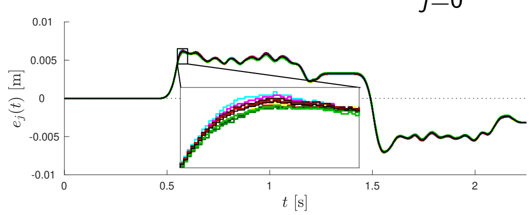
Traditional motion control(Oomen 2020*a*)



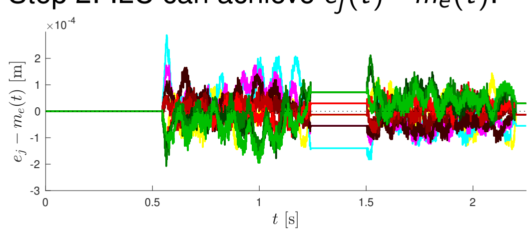
Highly reproducible error $e_j(t)$

Procedure(Oomen 2020*b*)

Step 1: compute $m_e(t) = \dfrac{1}{n_{\text{exp}}} \sum_{j=0}^{n_{\text{exp}-1}} e_j(t)$:



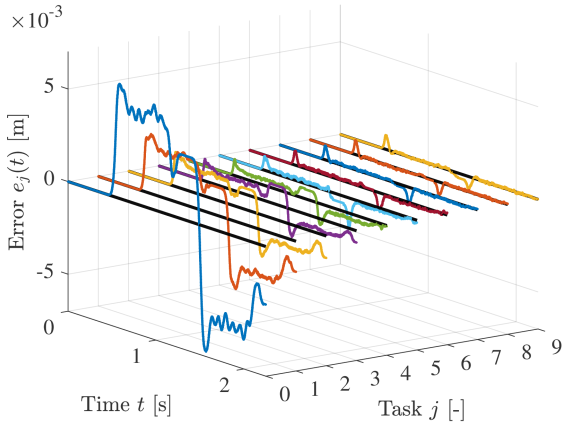
Step 2: ILC can achieve $e_j(t) - m_e(t)$:



How to achieve this?

Iterative Learning Control - What Can Be Achieved?

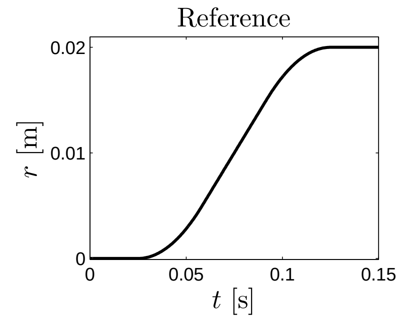
Spoiler



Traditional Motion Control: Servo Systems

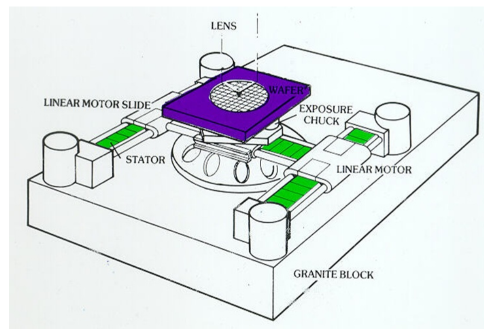
Servo control

- manipulate input such that output is close to reference
- e.g., motion systems



Control

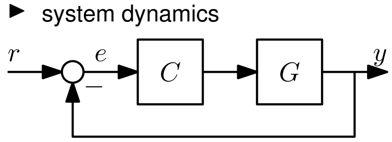
- feedback
- feedforward



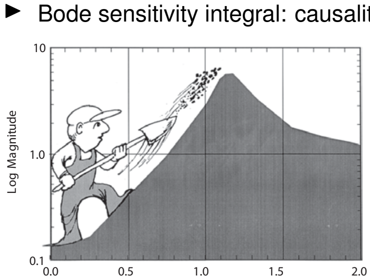
Traditional Motion Control: Feedback

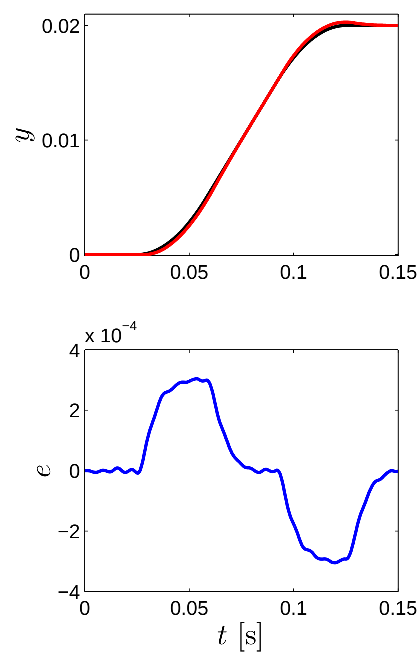
Feedback

- deal with uncertainty
  - disturbances
  - system dynamics



- performance limits
  - Bode sensitivity integral: causality

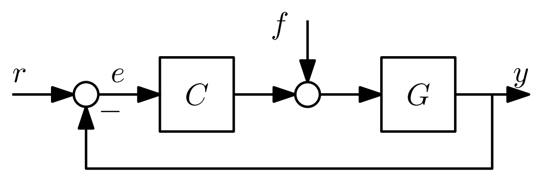




Traditional Motion Control: Feedforward

Feedforward

- compensate for *known* disturbances
  - e.g., reference
- Q: how to choose f?
- A: note that



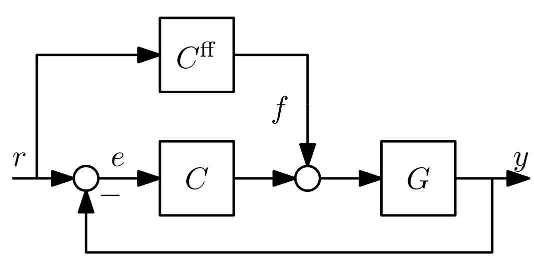
$$\begin{aligned} e &= r - Gf - GCe \\ (1+GC)e &= r - Gf \\ e &= S(r - Gf), \quad \text{with } S = (1+GC)^{-1} \end{aligned}$$

Thus: $f = G^{-1} r \Rightarrow e = 0$

Traditional Motion Control: Feedforward

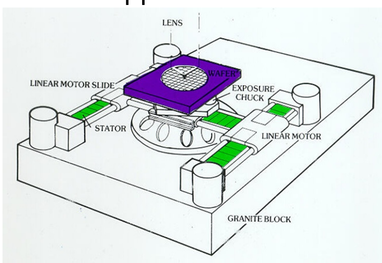
Feedforward

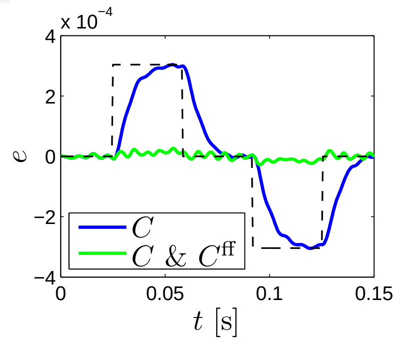
- often implemented as $f = C^{\mathrm{ff}} r$
- such that $C^{\mathrm{ff}} \approx G^{-1}$

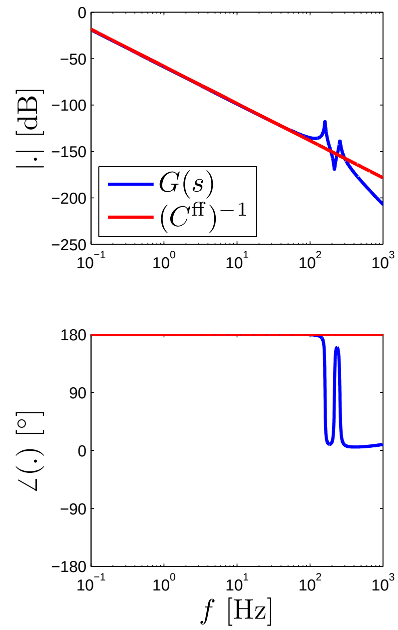


Traditional Motion Control: Example

Wafer stepper: ILC in (Roover & Bosgra 2000)



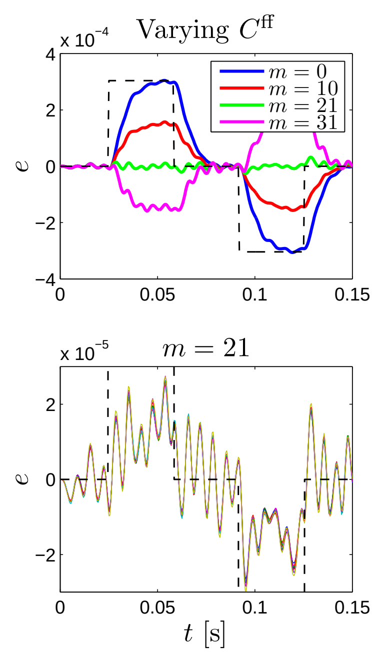




Traditional Motion Control: Example

Feedforward $f = C^{\mathrm{ff}} r$

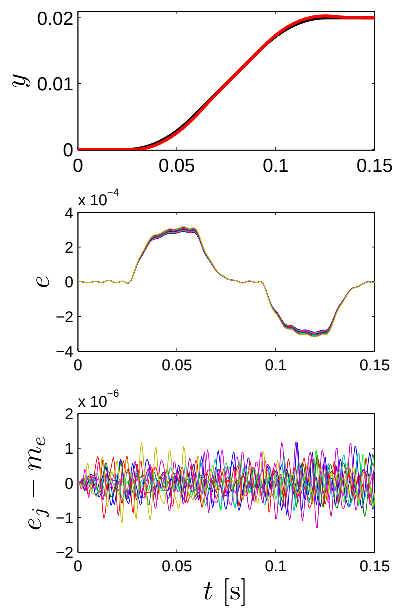
- compensates for reference r
  - requires knowledge of 'disturbance' r
  - applicable to large class of reference signals
- design strategy: $C^{\mathrm{ff}} \approx G^{-1} = ms^2$
  - model quality $\Rightarrow$ performance
  - optimal $m \Rightarrow$ repeating error
- let's analyse the error further...



Traditional Motion Control: Towards Learning

Applying the procedure of Slide 9

- for comparison: error without feedforward
- 10 Experiments
- highly reproducible error
- non-repeating part
  - $e_j - m_e, j = 1, \ldots, 10$
  - small



Next: ILC

- compensate the 'trial-invariant' disturbance that induces m_e

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ILC: Basic Principles

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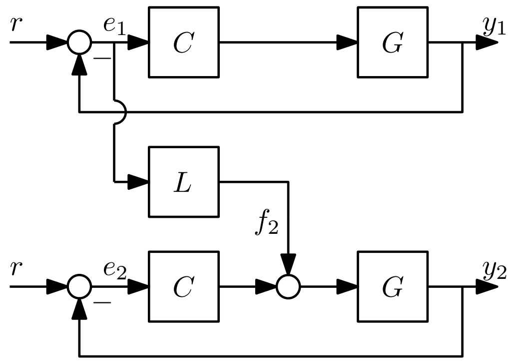

Experiment 1

- measure e_1
- assume r not measured

Experiment 2

- identical r
- which f_2 for small e_2?

ILC: Basic Principles

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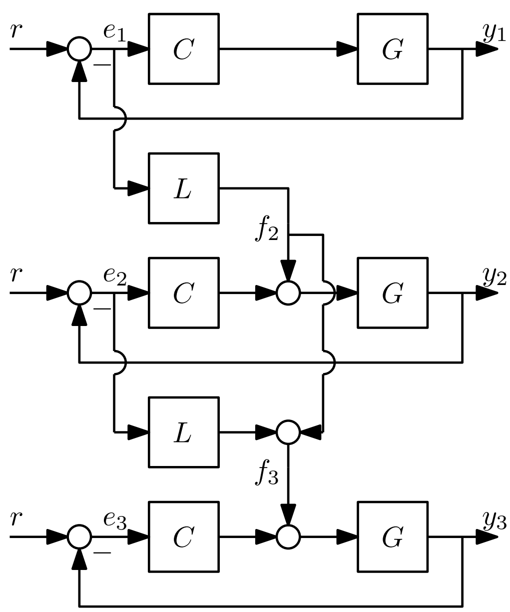


Experiment 2

- $e_2 = e_1 - GSLe_1$

Model errors

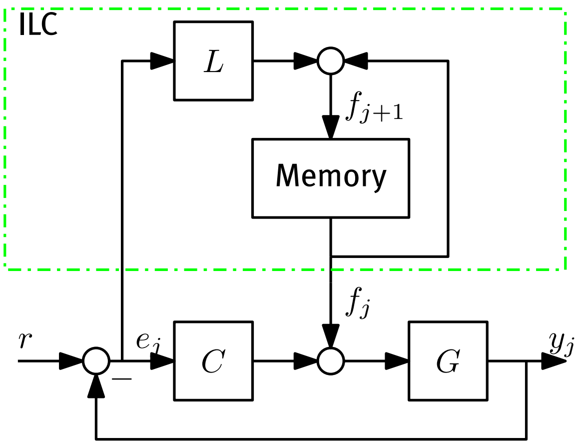
- in general: L^{-1} not exact
  $\Rightarrow e_2 \neq 0$

Idea

- do the same trick again!
- $f_{j+1} = \underbrace{f_j}_{\text{previous feedforward}} + \underbrace{Le_j}_{\text{correction}}$
- iterative learning!

ILC: Basic Principles

Basic ILC scheme



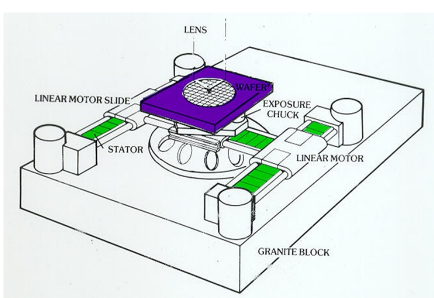
ILC algorithm

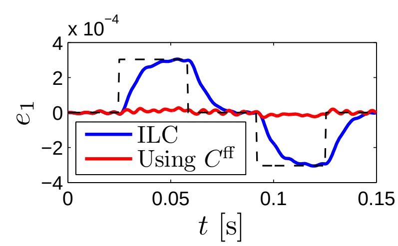
- learning update:
  - $f_{j+1} = f_j + Le_j$
  - compute off-line
- discrete time implementation:
  - memory: vector of data
  - ILC using
    - discrete time signals
    - discrete time systems

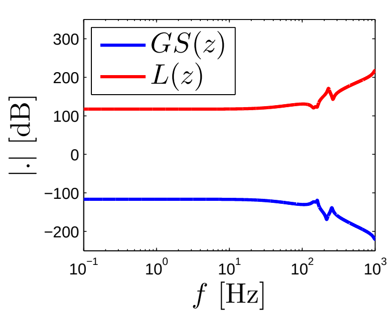
Next

- example

ILC: Basic Principles: Example



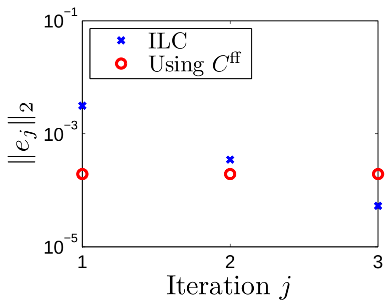


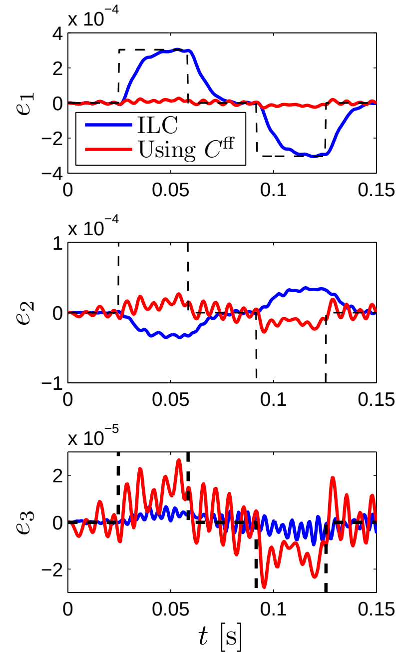


ILC: Basic Principles: Example

ILC iterations

1. Initial error
2. After one iteration:
   - error reduced
3. After two iterations:
   - error further reduced
   - outperforms C^{ff}!
   - more iterations?

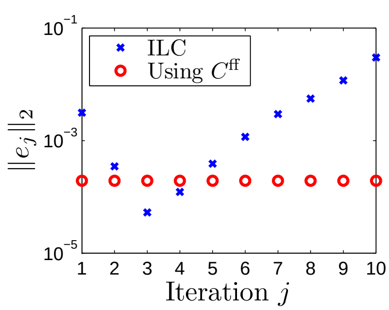


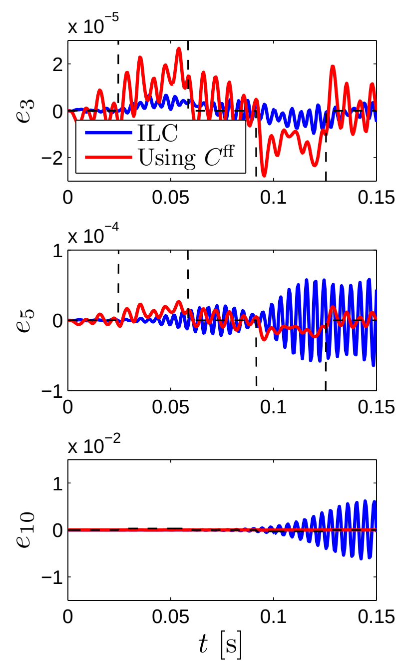


ILC: Basic Principles: Example

More ILC iterations

- increasing error in experiment 4, 5
- error increases for $j = 6, 7, \ldots 10$
- what's wrong?





Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

ILC: Basic Principles

Convergence Analysis

Enforcing Robustness

Implementation Aspects

Iteration-Variant Disturbances

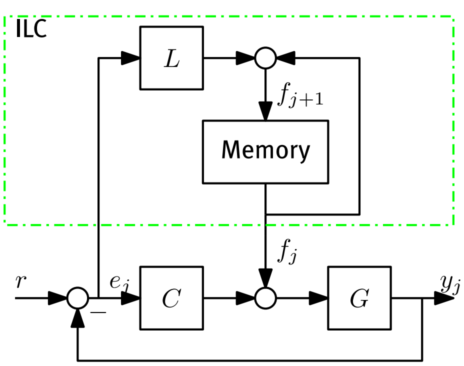
Alternative setup: serial ILC

Multivariable ILC

Summary

Convergence Analysis

Basic ILC scheme



ILC algorithm

- learning update:
  $f_{j+1} = f_j + Le_j$
- $e_j = Sr - GSf_j$
  $\Rightarrow f_j = (GS)^{-1}(Sr - e_j)$
- Error propagation:

$$\begin{aligned} e_{j+1} &= Sr - GSf_{j+1} \\ &= Sr - GS(f_j + Le_j) \\ &= (1 - GSL)e_j \end{aligned}$$

- Does the iterative scheme converge?

Convergence Analysis

ILC iteration

- $e_{j+1} = \underbrace{(1 - GSL)}_{=:F} e_j$

Approach

- infinite time $\Rightarrow$ frequency domain conditions
- based on contraction mapping
- underlying result:

  If F is a contraction, then the iteration $F(F(\ldots(Fe_j)\ldots))$ converges to a unique fixed point

Convergence Analysis

Theorem Let $f : X \mapsto X$, with $d(\cdot,\cdot)$ a metric on X. Then, f is called a contraction if there exists a $k \in [0, 1)$ such that

$$d(f(x_1), f(x_2)) \leq k d(x_1, x_2), \quad \forall x_1, x_2 \in X$$

Interpretation

- points $f(x_1)$ and $f(x_2)$ closer together than x_1 and x_2
- take as x_2 as $e_\infty = f(e_\infty)$ and x_1 as e_j. Then e_j converges to e_∞

Application to ILC

- use norms to define appropriate metric
- ℓ_2: square summable sequence, i.e., $\|x\|_2^2 = \sum_{i=-\infty}^{\infty} |x_i|^2 < \infty$
- then, $\|Fx_1 - Fx_2\|_2 = \|F(x_1 - x_2)\|_2 \leq \|F\|_{2,2}\|x_1 - x_2\|_2$

Key result: F is a contraction if $\|F\|_{2,2} < 1$

Convergence Analysis

Induced ℓ_2-norm

- $\|F\|_{2,2} = \|F\|_{\mathcal{L}_\infty} = \sup_{\omega \in [0,2\pi)} \left| F(e^{j\omega}) \right|$
- F need not be causal
  - see (Oomen & Rojas 2017) for a proof
  - will be essential in ILC: F is non-causal for typical non-causal L, $Q \Rightarrow$ unbounded $\mathcal{H}_\infty$ norm

A frequency domain ILC convergence check

The ILC iteration

$$e_{j+1} = \underbrace{(1 - GSL)}_{=:F} e_j$$

converges monotonically if

$$|1 - G(e^{j\omega})S(e^{j\omega})L(e^{j\omega})| < 1 \quad \forall \omega \in [0, 2\pi)$$

Convergence Analysis

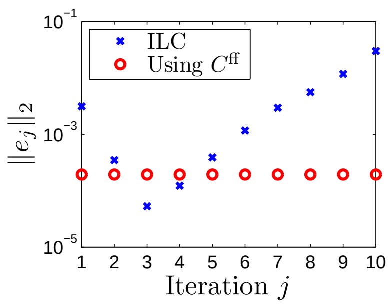
A simple example

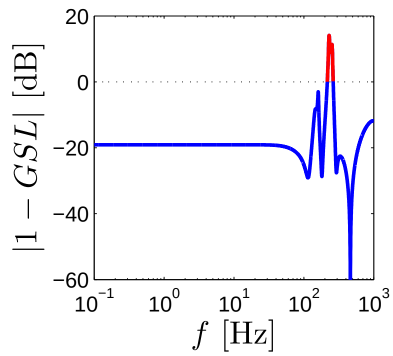
- error propagation: $e_{j+1} = (1 - GSL)e_j$
- convergence condition: $|1 - GSL| < 1 \quad \forall \omega \in [0, 2\pi)$
- consider static system:
  - $e_j \in \mathbb{R}, j = 1, 2, \ldots$
  - $GS = 4$
  - $L \in \mathbb{R}$

ILC iteration result

L	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$
$1 - GSL$			
e_1	16	16	16
e_2			
e_3			
e_4			
e_5			

Servo example again



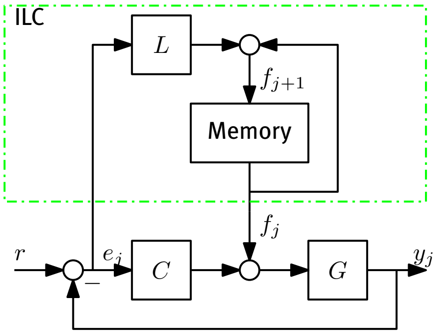


ILC convergence

- sampling time $h = 5 \cdot 10^{-4}$
- monotonic convergence if
  $|1 - G(e^{j\omega h})S(e^{j\omega h})L(e^{j\omega h})| < 1, \quad \forall \omega \in [0, 2\pi \cdot 2000)$
- condition violated in example

Enforcing Robustness

Basic ILC scheme



ILC algorithm

- learning update:

$$f_{j+1} = f_j + Le_j$$

- error propagation:

$$e_{j+1} = (1 - GSL)e_j$$

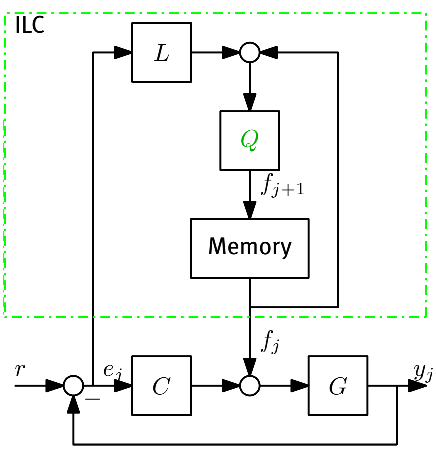
- monotonic convergence:

$$|1 - GSL| < 1, \quad \forall\omega \in [0, 2\pi) \tag{1}$$

- what if (1) is violated?

Enforcing Robustness

Basic ILC scheme



ILC algorithm

- introduce Q
- learning update:

$$f_{j+1} = Q(f_j + Le_j)$$

- error propagation:

$$e_{j+1} = Q(1 - GSL)e_j + (1 - Q)Sr$$

- monotonic convergence:

$$|Q(1 - GSL)| < 1, \quad \forall\omega \in [0, 2\pi) \tag{2}$$

- exploit freedom in Q to satisfy (2)!

Enforcing Robustness

Consequences of Q

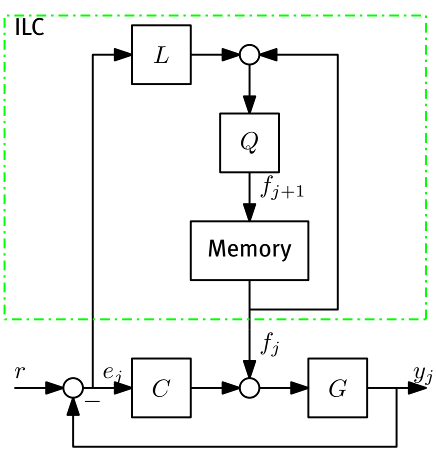
- after convergence:
  - $f_\infty := \lim_{j\to\infty} f_j, \quad e_\infty := \lim_{j\to\infty} e_j$
- learning update:
  $f_{j+1} = Q(f_j + Le_j) \quad \Rightarrow \quad f_\infty = \frac{QL}{1-Q} e_\infty$
- error at iteration j:
  $e_j = e_1 - GSf_j \quad \Rightarrow \quad e_\infty = \frac{1-Q}{1-Q(1-GSL)} e_1$

Result
Let $|Q(1-GSL)| < 1$. Then $Q = 1$ implies $e_\infty = 0$.

- implies $GSL \neq 0$
- if $|Q| < 1$, then e_∞ typically nonzero

Enforcing Robustness

ILC scheme



ILC design procedure

- learning update:

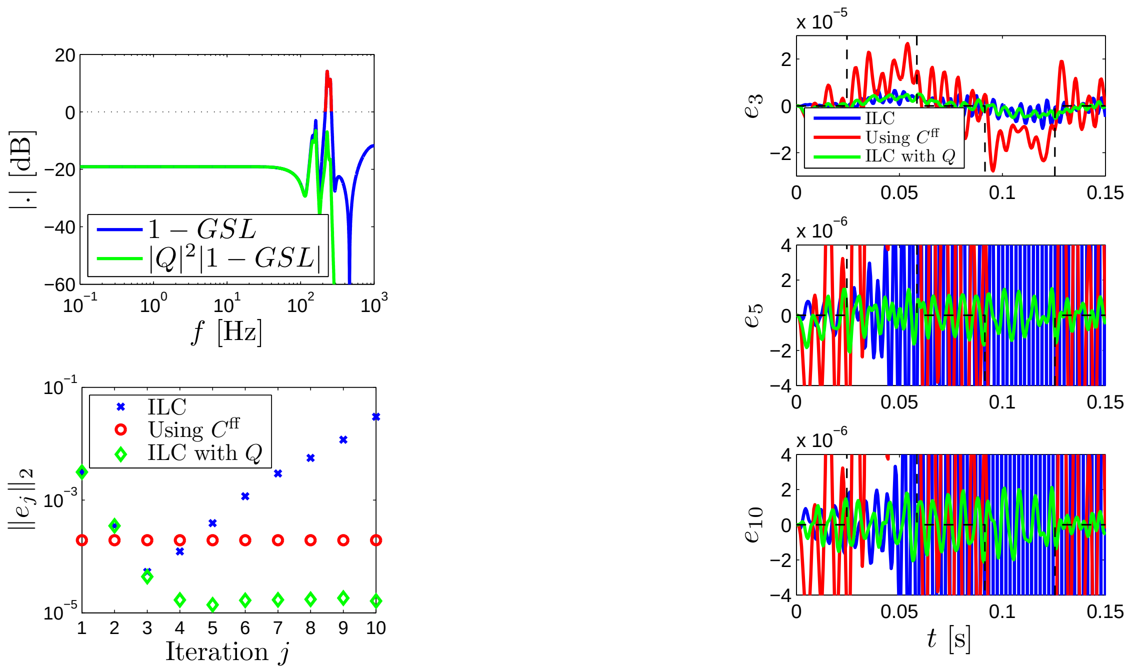
$$f_{j+1} = Q(f_j + Le_j)$$

1. design L to approximate $(GS)^{-1}$
2. design Q such that
   - for performance $Q(\omega) \approx 1$
   - for robustness

     $$|Q(1-GSL)| < 1, \quad \forall\omega \in [0, 2\pi)$$

     is satisfied
     - also using non-parametric FRF $GS(\omega)$

Enforcing Robustness: Example Revisited



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Alternative setup: serial ILC

Multivariable ILC

Summary

Implementation Aspects

ILC Design Requirements

- design L to approximate $(GS)^{-1}$
  - if GS nonminimum phase $\Rightarrow (GS)^{-1}$ unstable
  - if GS strictly proper $\Rightarrow (GS)^{-1}$ improper $\Rightarrow (GS)^{-1}$ non-causal
- design Q such that
  - $Q(\omega) \approx 1$ (performance)
  - $|Q(1 - GSL)| < 1 \; \forall \omega \in [0, 2\pi)$ (robustness)

Key reason that ILC leads to exceptional performance

- ILC anticipates on future disturbances
- achieved through design of non-causal L, Q
- enabled by off-line computations
- improved performance compared to feedback: causality!

Implementation Aspects

Approaches to compute L

A. ZPETC

B. Stable inversion

C. ...

many more optimal/nonoptimal finite/infinite preview approaches (van Zundert & Oomen 2018)

A. ZPETC

- Zero Phase Error Tracking Control(Tomizuka 1987)
- Let $GS(z^{-1}) = \frac{z^{-d}B(z^{-1})}{A(z^{-1})} = \frac{z^{-d}B_s(z^{-1})B_u(z^{-1})}{A(z^{-1})}$
  - z^{-d}: delay
  - $B_s(z^{-1})$: zeros strictly in $\mathbb{D}$
  - $B_u(z^{-1})$: zeros strictly outside $\mathbb{D}$
  - $\mathbb{D}$: open unit disc

Implementation Aspects

ZPETC approach

- If $GS(z^{-1}) = \frac{z^{-d}B_s(z^{-1})B_u(z^{-1})}{A(z^{-1})}$
- then $L(z^{-1}) = \frac{z^{d}A(z^{-1})B_u(z)}{\beta B_s(z^{-1})}$
- leads to $LGS = \frac{B_u(z)B_u(z^{-1})}{\beta}$

Key idea

- LGS has zero phase
- choose β such that $L(z^{-1})GS(z^{-1}) \approx 1$
  - e.g., $\beta = B_u(1)^2 \Rightarrow L = (GS)^{-1}$ for $\omega = 0$ ($z = e^{j\omega} = 1$)
- if $d = 0$ and $B_u(z^{-1}) = 1$, then $L = (GS)^{-1}$ (no error)

Implementation Aspects

ZPETC example

- $LGS = \frac{B_u(z)B_u(z^{-1})}{\beta}$
- let $B_u(z^{-1}) = b_0 + b_1 z^{-1}$
- then $B_u(z) = b_0 + b_1 z$
- and $B_u(z)B_u(z^{-1}) = b_0^2 + b_1^2 + b_0 b_1 z^{-1} + b_0 b_1 z$
- frequency response: substitute $z = e^{j\omega}$

$$\begin{aligned} B_u(e^{j\omega})B_u(e^{-j\omega}) &= b_0^2 + b_1^2 + b_0 b_1 e^{-j\omega} + b_0 b_1 e^{j\omega} \\ &= b_0^2 + b_1^2 + 2 b_0 b_1 cos(\omega) \in \mathbb{R} \end{aligned}$$

since $e^{j\omega} = \cos(\omega) + j\sin(\omega)$ (Euler)

Implementation Aspects

Implementing the ZPETC solution

- recall $L(z^{-1}) = \frac{z^d A(z^{-1}) B_u(z)}{\beta B_s(z^{-1})}$
- using $B_u(z) = z^p B_u^*(z^{-1})$
  - e.g., $B_u(z) = b_0 + b_1 z = z(b_1 + b_0 z^{-1}) = zB^*(z^{-1})$
- equivalent to

$$L(z^{-1}) = \underbrace{z^{p+d}}_{\text{anti-causal}} \underbrace{\frac{A(z^{-1}) B^*(z^{-1})}{\beta B_s(z^{-1})}}_{=:L_c(\text{causal})}$$

Matlab: `[n_Lc,d_Lc,phd] = ZPETC(n_GS,d_GS,1)`

- implement anti-causal z^{p+d} by applying a time shift, Matlab: for signal x, $x_{\text{shift}} = \left[x(phd+1), \ldots, x(N), 0_{phd}\right]^T$
- note: in finite time z^{p+d} and L_c do not commute (non-causal)

Implementation Aspects

B. Stable Inversion

- Let GS be square and have state-space realisation

$$\begin{aligned} x_0(k+1) &= A_0 x_0(k) + B_0 u(k) \\ y(k) &= C_0 x_0(k) + D_0 u(k) \end{aligned}$$

then $L = (GS)^{-1}$ is given by

$$\begin{aligned} x(k+1) &= \underbrace{A_0 - B_0 D_0^{-1} C_0}_{=:A} x(k) + \underbrace{B_0 D_0^{-1}}_{=:B} u(k) \\ y(k) &= \underbrace{-D_0^{-1} C_0}_{=:C} x(k) + \underbrace{D_0^{-1}}_{=:D} u(k) \end{aligned}$$

- if GS nonminimum phase, then L unstable
- if GS strictly proper, include additional unit step delays such that D_0 is invertible

Implementation Aspects

Solution through stable inversion

1. Introduce state transformation for L for dichotomy

$$\begin{bmatrix} x_+(k+1) \\ x_-(k+1) \end{bmatrix} = \begin{bmatrix} A_+ & 0 \\ 0 & A_- \end{bmatrix} \begin{bmatrix} x_+(k) \\ x_-(k) \end{bmatrix} + \begin{bmatrix} B_+ \\ B_- \end{bmatrix} y(k)$$

$$u(k) = \begin{bmatrix} C_+ & C_- \end{bmatrix} \begin{bmatrix} x_+(k) \\ x_-(k) \end{bmatrix} + Dy(k)$$

with $\lambda(A_+) \subset \mathbb{D}$ and $\lambda(A_-) \cap \mathbb{D} = \emptyset$

2. solve
$x_+(k+1) = A_+x_+(k) + B_+(k)y(k)$ forward in time with $x_+(0)$
$x_-(k+1) = A_-x_-(k) + B_-(k)y(k)$ backward in time with $x_-(N)$

3. compute $u(k) = \begin{bmatrix} C_+ & C_- \end{bmatrix} \begin{bmatrix} x_+(k) \\ x_-(k) \end{bmatrix} + Dy(k)$

Implementation Aspects

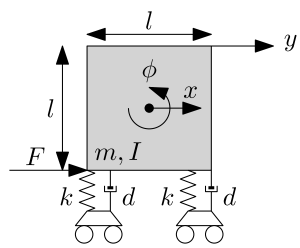
Remarks

- essentially bilateral Z/Laplace-transform instead of unilateral:
  L is seen as a non-causal instead of unstable (Vinnicombe 2001, Section 1.5)
- essentially solving a two-point boundary value problem
- extensions:
  - time-varying case (van Zundert & Oomen 2017*a*)
  - nonlinear case (Devasia et al. 1996)
- connections
  - lifted systems (Lunenburg 2010, Chapter 6)
  - LQ tracking (van Zundert, Bolder, Koekebakker & Oomen 2016)
- essential ingredient for rational feedforward (Blanken et al. 2017)
- Matlab: `stabsep`, `flipud`, `filter`
- next: simulation example

Implementation Aspects

Simulation example

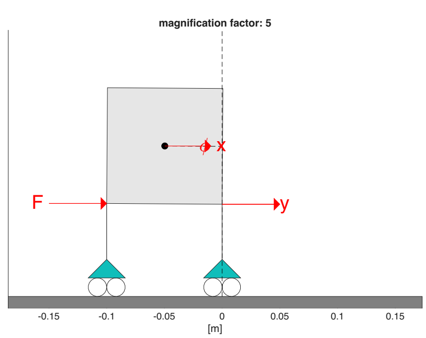
- flexible cart system
- varying sensor position
  - at bottom: collocated; minimum phase dynamics
  - at top: non-collocated; non-minimum phase dynamics
  - interpretation: depending on inertia/mass ratio, system either translates or rotates ‘first’ at sensor location
  - next: several movies illustrating consequences nonminimum phase dynamics
  - more details in (van Zundert & Oomen 2017*b*)

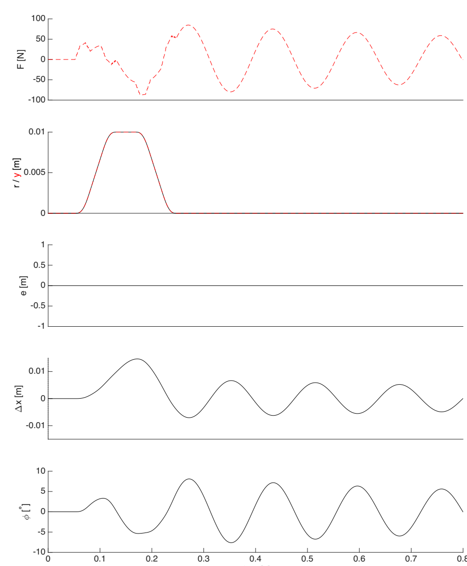


Implementation Aspects

Collocated system

- collocated
  - minimum phase
- approach
  - regular inverse
- result
  - perfect tracking
  - bounded input
  - causal





Implementation Aspects

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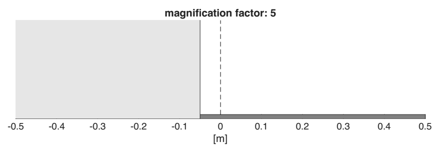
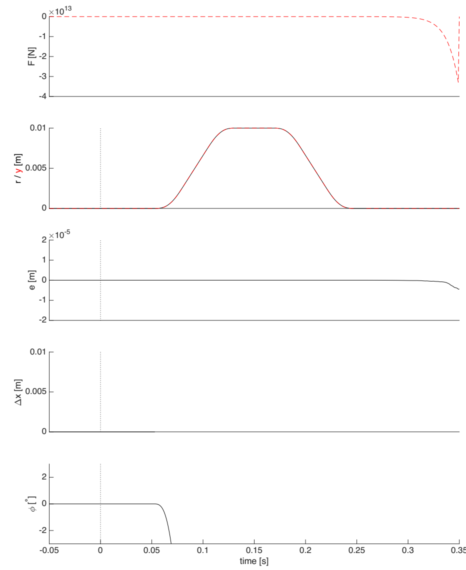
Non-collocated system

- non-collocated
  - non-minimum phase
- approach
  - regular inverse
- result
  - perfect tracking
  - unbounded input
  - causal



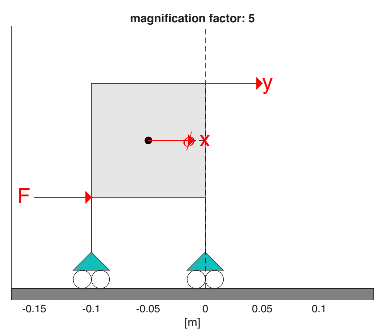


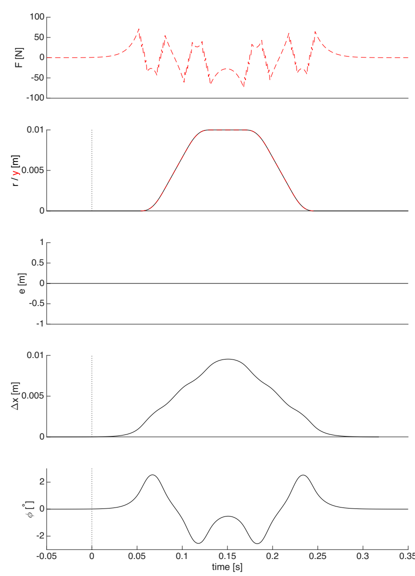
Implementation Aspects

46/62

Non-collocated system

- non-collocated
  - non-minimum phase
- approach
  - stable inverse
- result
  - perfect tracking
  - bounded input
  - noncausal





Implementation Aspects

Design non-causal Q filter

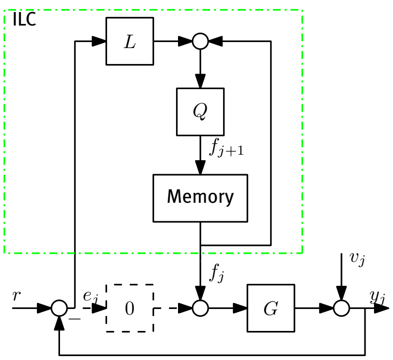
- $Q = 1$: perfect tracking
- $Q \neq 1$: nonperfect tracking
  - $|Q| \neq 1$
  - $\angle(Q) \neq 0$
- mitigate effect of $\angle(Q) \neq 0$ by using adjoint $Q^*(z) = Q(\frac{1}{z})$
  - Q: forwards in time
  - Q^*: backwards in time
  - $Q^*(e^{j\omega})Q(e^{j\omega}) = Q(e^{-j\omega})Q(e^{j\omega}) = |Q(e^{j\omega})|^2 \Rightarrow$ no phase shift
  - monotonic convergence: $|Q|^2|(1 - GSL)| < 1, \quad \forall \omega \in [0, 2\pi)$
  - Matlab: `filtfilt`

Iteration-Variant Disturbances

Disturbance properties

- tacit assumption: disturbances iteration invariant
- what if disturbances are iteration variant (Oomen & Rojas 2017)

ILC scheme



Similar setup

- v_j: iteration variant disturbance
- r: iteration invariant 'disturbance'
- $G \in \mathcal{RH}_\infty$, $C = 0 \Rightarrow S = 1$
- $L, Q \in \mathcal{RH}_\infty$

ILC equations

- $e_j = Sr - Sv_j - GSf_j$
- $$\begin{aligned} f_{j+1} &= Q(f_j + Le_j) \\ &= Q((1 - LG)f_j + Lr - Lv_j) \end{aligned}$$

Iteration-Variant Disturbances

Why is this challenging to analyse?

- $f_{j+1} = Q((1 - LG)f_j + Lr - Lv_j)$
- initialization: $f_1 = 0$

Successive substitution

j=1. $f_2 = Q(Lr - Lv_1)$

j=2. $$\begin{aligned} f_3 &= Q((1 - LG)f_2 + Lr - Lv_2) \\ &= (Q(1 - LG) + 1)QLr - Q(1 - LG)QLv_1 - QLv_2 \end{aligned}$$

j. $$f_{j+1} = \underbrace{\sum_{i=1}^{j} (Q(1 - LG))^{i-1} QLr}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^{j} (Q(1 - LG))^{j-n} QLv_n}_{\text{all past noise realisations...!}}$$

Iteration-Variant Disturbances

Why is this challenging to analyse?

- at iteration j: $f_{j+1} = \underbrace{\sum_{i=1}^{j} (Q(1-LG))^{i-1} QLr}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^{j} (Q(1-LG))^{j-n} QLv_n}_{\text{all past noise realisations...!}}$

Result (Oomen & Rojas 2017)

- Assume v_j has iteration invariant properties
  - $v_j = Hn_j$, n_j i.i.d. ZMWN, variance λ_e
  - H monic, bistable
  - leads to spectrum $\phi_v = |H|^2 \lambda_e$
- Assume $|Q(1-GL)| < 1$ (convergence)

Then, for $j \to \infty$, $\quad \phi_{e_\infty} = \left| \frac{1-Q}{1-Q(1-LG)} \right|^2 \phi_r + \left(1 + \frac{|GQL|^2}{1-|Q(1-LG)|^2} \right) \phi_v$

Iteration-Variant Disturbances

Result

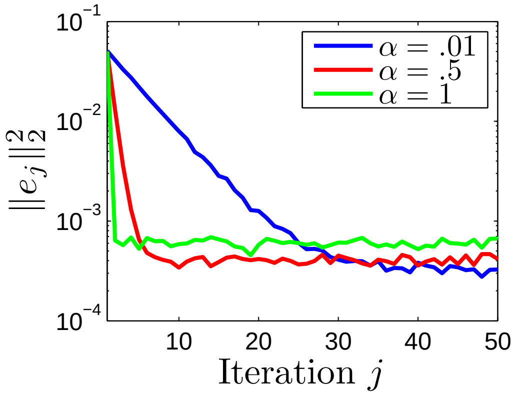
$$\phi_{e_\infty} = \left| \frac{1-Q}{1-Q(1-LG)} \right|^2 \phi_r + \left(1 + \frac{|GQL|^2}{1-|Q(1-LG)|^2} \right) \phi_v$$

Interpretation - 1

- assume $Q = 1$, $L = G^{-1}$
- $\phi_{e_\infty} = (1 + \frac{1}{1})\phi_v = 2\phi_v$
- learning amplifies iteration-variant disturbances!

Interpretation - 2

- assume $Q = 1$, $L = \alpha G^{-1}$
- $\phi_{e_\infty} = (1 + \frac{\alpha^2}{1-|1-\alpha|^2})\phi_v = (1 + \frac{\alpha^2}{-\alpha^2+2\alpha})\phi_v$
- for small α: $\phi_{e_\infty} \approx (1 + \frac{1}{2}\alpha)\phi_v$
- decreasing $\alpha \Rightarrow$ reducing asymptotic error



Reducing α

- reduced learning speed
- reduced amplification trial-variant disturbances

Example

- $f_{j+1} = f_j + Le_j$
- $L = \alpha(GS)^{-1}$
- $\alpha = 0.01, 0.5, 1$

Steady state error

α	$\frac{1}{N}\sum_{j=n+1}^{n+N} \|e_j\|_2^2$
0.01	$0.32 \cdot 10^{-3}$
0.5	$0.40 \cdot 10^{-3}$
1	$0.60 \cdot 10^{-3}$

Trade-off iteration-(in)variant disturbances for case $L = \alpha(GS)^{-1}$, $Q = 1$

- learning speed w.r.t. iteration invariant disturbances
- amplification of iteration variant disturbances
- remark: if $Q \neq 1$, use analysis of Slide 32

Extended ILC algorithms for iteration-variant disturbances

- high order ILC: $f_{j+1}(f_j, e_j, e_{j-1}, \ldots)$
- stochastic approximation
  - $f_{j+1} = Q(f_j + \gamma_j Le_j)$
  - $\gamma_j \in \mathbb{R}_+$ decreasing sequence
- sparse ILC(Oomen & Rojas 2017)

Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

ILC: Basic Principles

Convergence Analysis

Enforcing Robustness

Implementation Aspects

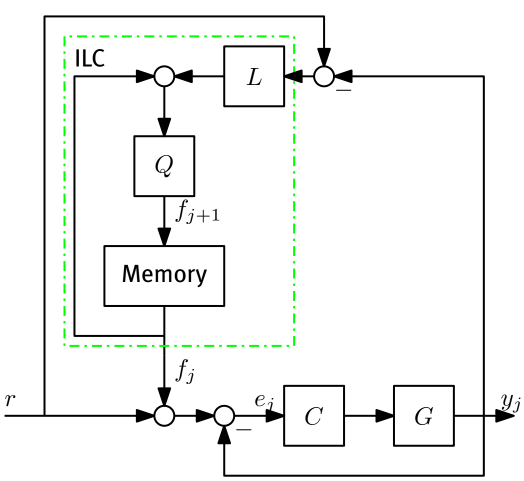
Iteration-Variant Disturbances

Alternative setup: serial ILC

Multivariable ILC

Summary

Alternative setup: serial ILC



Alternative implementation

- serial instead of parallel ILC
- similar theory
- suitable for dedicated hardware with only access to reference signal
- sometimes advantageous if ILC measured signal is not equal to feedback measured signal
  (Bolder & Oomen 2016)

Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

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Alternative setup: serial ILC

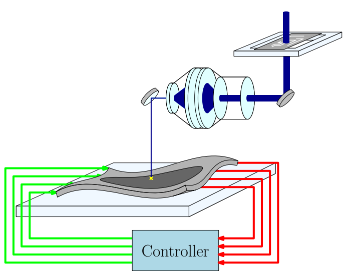
Multivariable ILC

Summary

Multivariable ILC

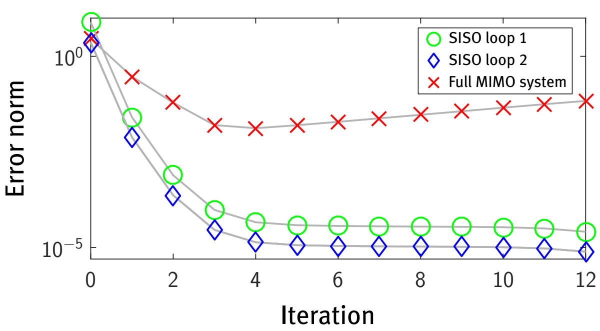
Multivariable ILC

- future mechatronic systems have many inputs and outputs
- tempting: use well-known SISO techniques to learn loop-by-loop



✗ This does not work!

What's wrong?



Multivariable ILC

What's wrong? interaction!

- much worse for ILC: 'bandwidth' up to Nyquist frequency

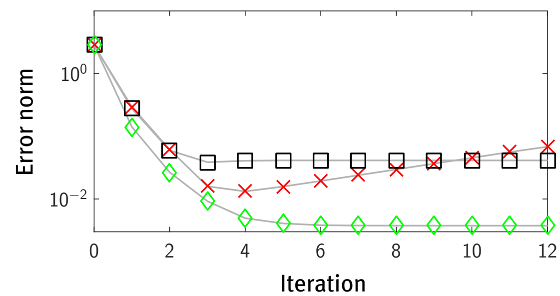
Multivariable ILC design (Blanken & Oomen 2020)

	Performance	Modeling
SISO ignore	no	SISO
Decentralized	good	SISO
Centralized	excellent	MIMO

Recall: feedback design procedure(Oomen 2018*a*)

1. Traditional: ignore interaction
2. Static decoupling
3. Manual design for interaction
4. Optimal & robust control

Balances modeling cost vs. control performance



Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

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Summary

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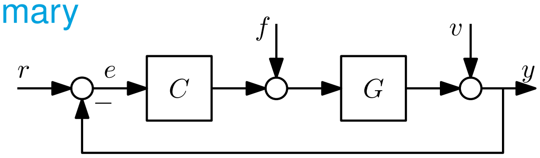
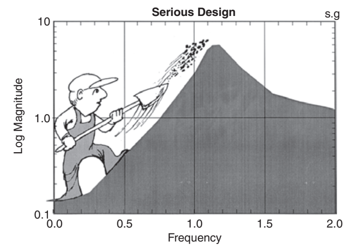
Summary





- $e = S(r - Gf) + Sv$
- feedback:
  - deal with non-repetitive disturbances
  - attenuate v: if $v = He$, e ZMWN, then $S \approx H^{-1}$ (Boeren et al. 2017) (van Zundert & Oomen 2017*c*)
- setpoint generator: select r, e.g., for point-to-point motion (Lambrechts et al. 2005) (Boeren et al. 2014)
- feedforward:
  - determine θ such that $f = F(\theta)r$, with $F(\theta) \approx G^{-1}$ (parameterized feedforward)
  - determine f such that $f \approx G^{-1}r$ via iteration-domain feedback (ILC) (all repetitive disturbances)

Summary

Summary

- basic ILC principle introduced
- convergence analysis:
  - model errors: need for robustness
  - design variables: L and Q
- iteration invariant vs. iteration variant disturbances
- developed framework
  - infinite time
  - L, Q: non-causal LTI filters
  - frequency domain/loopshaping design techniques

Summary

Reading

- ILC design procedure from these slides: (Oomen 2020*b*) (Blanken et al. 2019)
- general background on learning: (Oomen 2018*b*)
- inversion techniques overview: (van Zundert & Oomen 2018)
- noise analysis: (Oomen & Rojas 2017)
- general ILC overview paper: (Bristow et al. 2006)
- basics motion feedback and feedforward: (Oomen 2020*a*)
- advanced motion control developments: (Oomen 2018*a*)

Summary

Outlook

- finite time ILC analysis and synthesis
- L, Q: non-causal LTV filters
- and its extensions
  - LTV/position-dependent ILC (van Zundert, Bolder, Koekebakker & Oomen 2016)
  - resource-efficient ILC (van Zundert, Bolder, Koekebakker & Oomen 2016)
  - ILC for for varying tasks (van Zundert, Bolder & Oomen 2016) (Bolder & Oomen 2015)

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